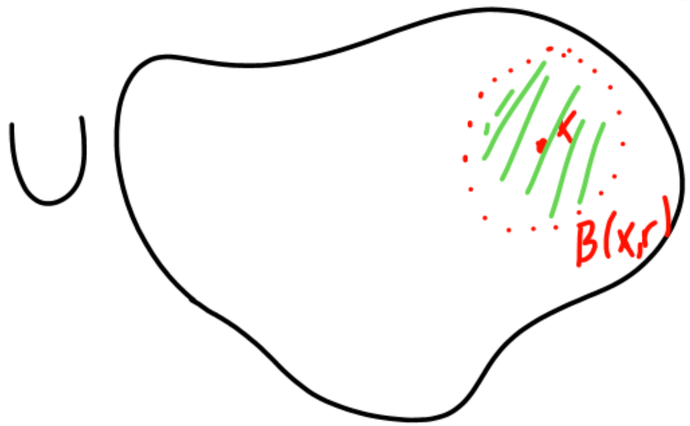
Laplace's Equation

$$-\Delta u = 0$$

If u solves this, we say u is <u>harmonic</u>.

Harmonic Functions have lots of good properties.



Thm (Mean Value Property)

If $u \in C^2(U)$ for some open domain U, then for any ball $B(x,r)$ with $\overline{B(x,r)} \subset U$,

$$\boxed{u(x) = \fint_{\partial B(x,r)} u(y)\, dS(y) = \fint_{B(x,r)} u(y)\, dy.}$$

<u>Pf</u> Define

$$\phi(r) = \fint_{\partial B(x,r)} u(y)\, dS(y)$$

Idea: Diff. ϕ in r, show

$$\phi'(r) = 0.$$

$$\phi(r) = \frac{1}{n\alpha(n) r^{n-1}} \int_{\partial B(x,r)} u(y)\, dS(y)$$

Let $y = x + rz$, $z = \frac{y-x}{r}$

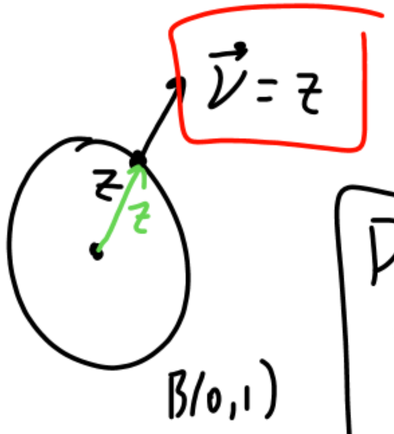
$$dS(y) = \boxed{r^{n-1}}\, dS(z).$$

Then

$$\phi(r) = \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} u(x+rz)\, dS(z)$$

$$\frac{d}{dr}\phi(r) = \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} \boxed{\frac{d}{dr} u(x+rz)}\, dS(z)$$

$$= \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} \nabla u(x+rz) \cdot z\, dS(z)$$



$$= \frac{1}{n\alpha(n)} \int_{B(0,1)} \nabla \cdot (\nabla u(x+rz))\, dz$$

Div Thm:

$$\int_{\partial U} F \cdot \vec{\nu}\, dS = \int_U \nabla \cdot F\, dy$$

Recall:

$$B(x,r) = \{ y \in \mathbb{R}^n : |y-x| < r \}$$

$$\overline{B(x,r)} = \{ y \in \mathbb{R}^n : |y-x| \leq r \}$$

$$\nabla \cdot (\nabla f) = \Delta f$$

$$\phi'(r) = \frac{1}{n\alpha(n)} \int_{B(0,1)} \nabla \cdot \left(\nabla u(x+rz) \right) dz$$

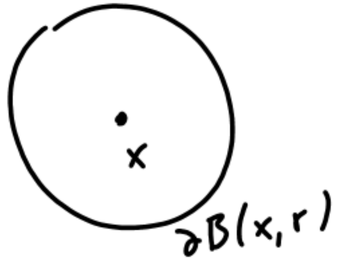
$$= \frac{1}{n\alpha(n)} \int_{B(0,1)} \Delta u(x+rz)\, dz$$

$$= 0 \quad \text{since } \Delta u = 0.$$

$\Rightarrow \phi$ is constant in r.

Since u is continuous,

$$\phi(r) = \fint_{\partial B(x,r)} u(y)\, dS(y) \longrightarrow u(x) \quad \text{as } r \to 0$$



$\Rightarrow \phi(r) = u(x)$ for all $r > 0$ with $\overline{B(x,r)} \subset U$

To prove $u(x) = \fint_{B(x,r)} u(y)\,dy$,

integrate

$$u(x) = \fint_{\partial B(x,r)} u(y)\,dS(y)$$

from 0 to R.

$$\int_0^R u(x)\,dr = \int_0^R \frac{1}{n\alpha(n) r^{n-1}} \int_{\partial B(x,r)} u(y)\,dS(y)\,dr$$

$$R u(x) = \int_0^R \int_{\partial B(x,r)}$$

$$r^{n-1} u(x) = \frac{1}{n\alpha(n)} \int_{\partial B(x,r)} u(y)\,dS(y)$$

Integrate from 0 to R:

$$\int_0^R r^{n-1} u(x)\,dr = \frac{1}{n\alpha(n)} \int_0^R \int_{\partial B(x,r)} u(y)\,dS(y)\,dr$$

$$\frac{R^n}{n} u(x) = \frac{1}{n\alpha(n)} \int_{B(x,r)} u(y)\,dy$$

$$u(x) = \frac{1}{\alpha(n) R^n} \int_{B(x,r)} u(y)\,dy = \fint_{B(x,R)} u\,dy$$

Maximum Principle

Thm Let $u \in C^2(U)$ be harmonic. Then

a) $\max_{\overline{U}} u = \max_{\partial U} u$

b) If $u(x_0) = \max_{\overline{U}} u$ at some interior point x_0, then u is constant.

Strong Maximum Principle.

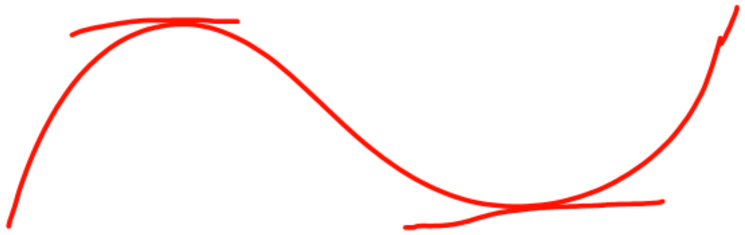
$\overline{U} = U \cup \partial U$

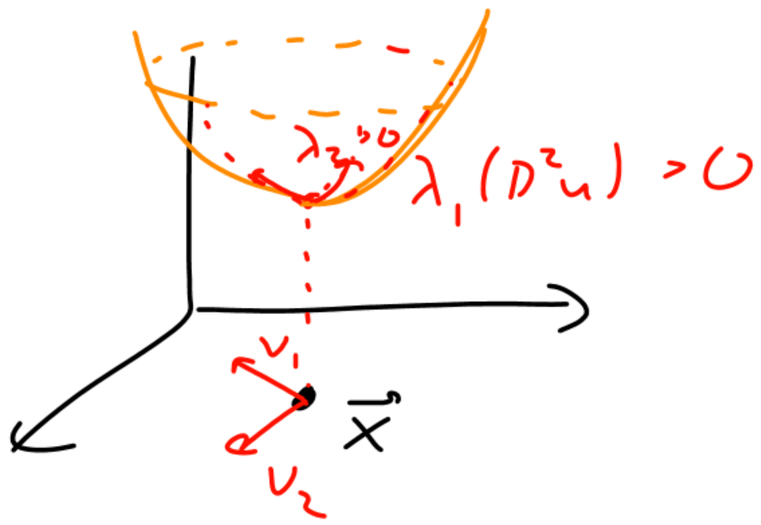
Pf Idea:

$$0 = \Delta u = \sum_{i=1}^{n} u_{x_i x_i}$$

$$= \operatorname{tr}(D^2 u)$$

$$\left((D^2 u)_{ij} = u_{x_i x_j} \right) \quad \hookrightarrow = \sum_{j=1}^{n} \lambda_j (D^2 u)$$





Idea: If u has a <u>strict</u> local max at x_0, then all $\lambda_j(D^2u) < 0$ at x_0, contradicting $\Delta u = 0$.

Q: How to rule out non-strict local maxima?

Let $\varepsilon > 0$ be a small number, define $u_\varepsilon(x) = u(x) + \varepsilon|x|^2$

Claim: $u_\varepsilon(x)$ cannot have an interior maximum point (strict or not). Assume x_0 is local max, then

$$\Delta u_\varepsilon = tr(D^2u_\varepsilon) = \sum_{j=1}^{n} \lambda_j(D^2u_\varepsilon) \leq 0$$

$\Delta u_\varepsilon \le 0$, but

$$\Delta u_\varepsilon = \Delta\left(u + \varepsilon |x|^2\right)$$
$$= \Delta u + \Delta\left(\varepsilon |x|^2\right)$$
$$= 0 + \sum_{i=1}^{n} \varepsilon \frac{\partial^2}{\partial x_i^2}\left[|x|^2\right]$$

$$|x|^2 = \sum_{k=1}^{n} x_k^2$$

$$\frac{\partial}{\partial x_i} |x|^2 = \frac{\partial}{\partial x_i}\left(\sum_{k=1}^{n} x_k^2\right) = 2x_i$$

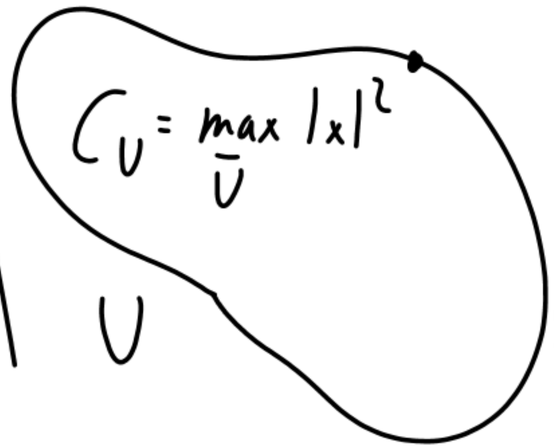
$$\frac{\partial^2}{\partial x_i^2} |x|^2 = 2$$

$$\Delta u_\varepsilon = 2n\varepsilon > 0.$$

$\Rightarrow$ u_ε can't have an interior maximum

$$\Rightarrow \max_{\bar{U}} u_\varepsilon = \max_{\partial U} u_\varepsilon$$

$$\max_{\bar{U}} u \le \max_{\bar{U}} \left(u + \varepsilon |x|^2\right) = \max_{\partial U} \left(u + \varepsilon |x|^2\right)$$
$$\le \max_{\partial U} u + \varepsilon C_U$$



Send $\varepsilon \to 0$:

$$\boxed{\max_{\bar{U}} u \le \max_{\partial U} u}$$

· Harmonic functions also satisfy "minimum principle", since $-u$ is also harmonic

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Using this, we can prove uniqueness theorem for

$$(*)\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U \end{cases}$$

<u>Thm</u> There is at most one solution to $(*)$.

<u>Pf</u> Let u, v be two solutions to $(*)$.

Let $w = u - v$.

$$-\Delta w = -\Delta(u-v) = -\Delta u + \Delta v = f - f = 0$$

On ∂U, $w(x) = u(x) - v(x) = g(x) - g(x) = 0$

We claim $w = 0$ in U.

Max Principle:

$$\max_{\bar{U}} w = \max_{\partial U} w$$
$$= 0$$
$$\Rightarrow w \leq 0 \text{ in } U.$$

Minimum principle:

$$\min_{\bar{U}} w = \min_{\partial U} w = 0$$
$$\Rightarrow w \geq 0 \text{ in } U$$

$\Rightarrow w = 0$, so $u = v$.

<u>Remark</u>: The last proof worked because w solved $-\Delta w = 0 \Rightarrow$ Max. Principle

However, solutions of Poisson's Eqn $-\Delta u = f$ <u>don't</u> satisfy a maximum principle.

<u>Positivity</u>:

Consider

$$\begin{cases} -\Delta u = 0 & \text{in } U \\ u = g & \text{on } \partial U \end{cases}$$

If $g > 0$ everywhere, then $u > 0$ in U.

(by Minimum Principle)

Next goal: Prove Harmonic functions are smooth (C^∞)

Tool: Mollification.

The <u>standard mollifier</u> η is a function on $\mathbb{R}^n$ with:

- $\eta \geq 0$,
- η is C^∞
- $\int_{\mathbb{R}^n} \eta(x)\,dx = 1$

and

- $\eta(x) = 0$ outside $B(0,1)$

e.g. $\eta(x) = \begin{cases} C \exp\left(\frac{1}{|x|^2 - 1}\right), & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$